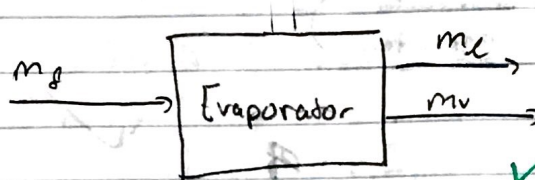


CHE 210 HW #3

1a)



- 1) 10
- 2) 20
- 3) ~~15~~ 20
- 4) ~~15~~ 15
- 5) 20

10/20

b)

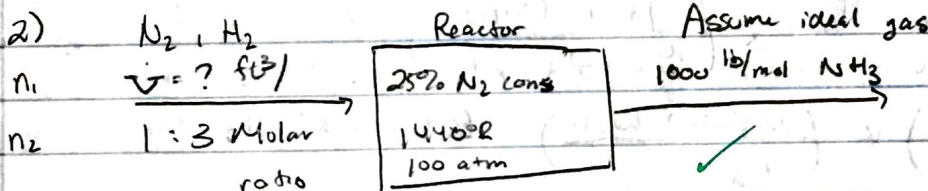
85% benzene + 15% toluene
10.0% benzene + 89.4% toluene

5) 20

c) underspecified

85/100

2)



$$Acc = In - out + Gen - Con$$

$$N_{in} - N_{cons} - N_{out} = 0$$

$$H_{in} - H_{out} - H_{con}$$

$$N_{in} - N_{cons} = N_{out}$$

$$H_{in} - H_{cons} = H_{out}$$

$$In: N_{in}/H_{in} = \frac{1}{3} \quad N_{cons} = 0.25 N_{in} \quad Accumulation / Neon = 2$$

$$N_{cons} = \frac{1000 \text{ lbmol/h}}{2} = 500 \text{ lbmol/h}$$

$$N_{cons} = 500 \text{ lbmol/h}$$

$$N_{in} = N_{cons} / 0.25 = 2000 \text{ lbmol/h}$$

$$H_{in} / N_{in} = 3 \quad H_{in} = 6000 \text{ lbmol/h}$$

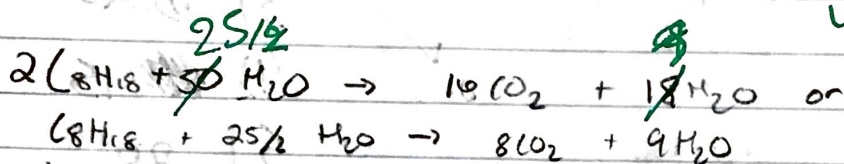
$$Total \text{ in} = 2000 + 6000 = 8000 \text{ lbmol/h} \left(\frac{0.732 \text{ atm ft}^3}{\text{lbmol}^\circ R} \right) \left(\frac{1440^\circ R}{100 \text{ atm}} \right) = 84119.04 \text{ ft}^3/\text{h}$$

20/20

4) 5.6 billion metric tons CO_2 about 0.25

300 million people CO_2 per person

$$\frac{(0.25)(5.6 \text{ billion})}{300 \text{ million}} \times \left(\frac{2204.62}{\text{metric ton}} \right) = 10,288.23 \text{ } \text{CO}_2 \text{ lb/per person}$$



13,476 mi/yr according to .gov

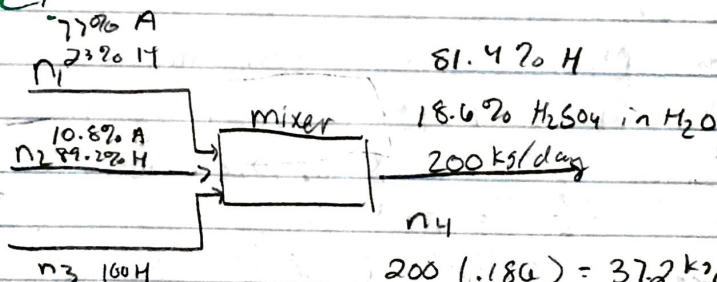
15/20

$$\frac{13,476 \text{ mi}}{\text{yr}} \times \left(\frac{1 \text{ gal}}{27.5 \text{ mi}} \right) \times \left(\frac{0.00378541 \text{ m}^3}{1 \text{ gal}} \right) \times \left(\frac{100 \text{ cm}}{1 \text{ m}} \right)^3 \times \left(\frac{1 \text{ mL}}{1 \text{ cm}^3} \right) \times \left(\frac{0.7 \text{ g } (\text{C}_8\text{H}_{18})}{1 \text{ mL}} \right) \times \left(\frac{1 \text{ mol } (\text{C}_8\text{H}_{18})}{144.232 \text{ g } (\text{C}_8\text{H}_{18})} \right) \times \left(\frac{8 \text{ mol } (\text{CO}_2)}{1 \text{ mol } (\text{C}_8\text{H}_{18})} \right) \times \left(\frac{43.99 \text{ g } (\text{CO}_2)}{1 \text{ mol } (\text{CO}_2)} \right) \times \left(\frac{0.00220462 \text{ lb}}{1 \text{ g}} \right) = 6938.95 \text{ lb } \text{CO}_2$$

Is it reasonable?

5) A = H_2SO_4

H = H_2O



$$200 (1.86) = 37.2 \text{ kg/day}$$

Assume steady state

$$\text{DOF} = 7 - 6 = 1$$

Accum = In - out + Gen - cons

$$\text{Aid: } .77 n_1 + .108 n_2 - .186 n_4 + 0 - 0 = 0$$

$$.77 n_1 + .108 n_2 = 37.2 \text{ kg/day}$$

$$\text{H}_2\text{O: } .23 n_1 + .892 n_2 - .814 (n_4) = 0$$

$$n_3 + .23 n_1 + .892 = 162.8 \text{ kg}$$

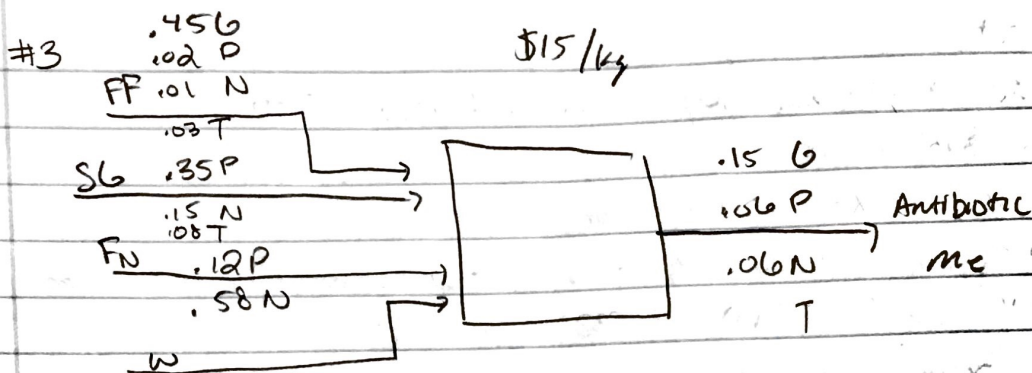
not specified enough.

$$n_2 = 0 \quad n_1 = 48.3 \text{ kg/day} \quad n_3 = 151.7 \text{ kg/day}$$

$$= \$4.83 \quad = \$3.034$$

$$= \$7.86$$

20/20



$$G: .45 FF + 0 SG + 0 FN = .15$$

$$P: .02 FF + .35 SG + .12 FN = .06$$

$$N: .01 FF + .15 SG + .58 FN = .06$$

$$\begin{bmatrix} .45 & 0 & 0 \\ .02 & .35 & .12 \\ .01 & .15 & .58 \end{bmatrix} \begin{bmatrix} FF \\ SG \\ FN \end{bmatrix} = \begin{bmatrix} .15 \\ .06 \\ .06 \end{bmatrix}$$

$$(FF = .333 \text{ kg}, SG = .130 \text{ kg}, FN = .004 \text{ kg}) \text{ per batch}$$

$$w: m_{FF} + m_{SG} + m_{FN} + m_w = m_e$$

$$m_w = .472 \text{ kg / batch}$$

b) 20 kg batch cost \$2000

$$.333 \text{ kg} \times \frac{\$20}{\text{kg}} + .130 \text{ kg} \times \frac{\$10}{\text{kg}} + .004 \text{ kg} \times \frac{\$15}{\text{kg}} = \$8.92$$

$$\$15 - \$8.92 = \$6.08 / \text{kg} = \text{savings}$$

$$\$6.08 (20) = \$121.6 \quad 2000 / 121.6 = 16.45 \text{ batches}$$

$$\approx 17 \text{ batches for savings}$$

$$\frac{1520}{20}$$

~~1520~~

c) $I_n = out$

$$.03(.333) + .08(.130) = .0204 (100\%) = 2.04\%$$

$$.03(.333) + .08(x) = .03$$

$$x = .25 \text{ kg for } 32 \text{ wt}$$

with 3wt% we would have to increase Super Grow since its cheaper.

$$(.333)(20) + (.25)(10) + (.033)(15) = \$9.66$$

the price would increase

